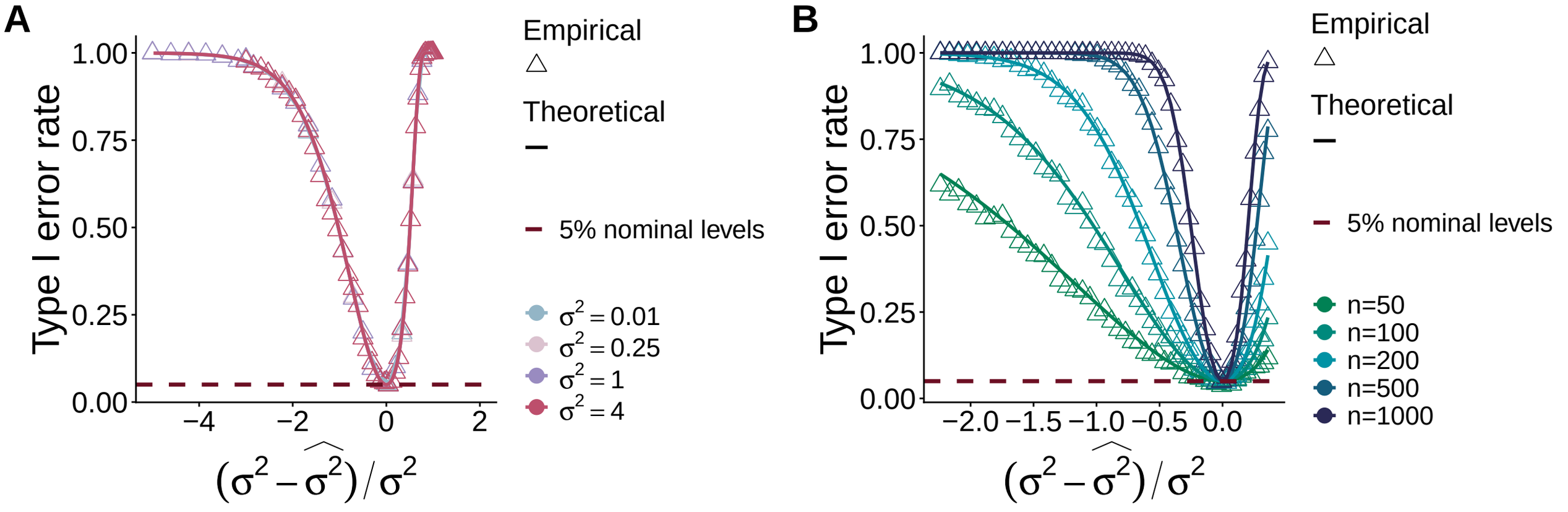
A
B
Type I error rate
1.00
0.75
0.50
0.25
0.00
−4
−2
0
2
−2.0
−1.5
−1.0
−0.5
0.0
$(\sigma^2 - \widehat{\sigma^2})/\sigma^2$
Empirical
△
Theoretical
5% nominal levels
$\sigma^2 = 0.01$
$\sigma^2 = 0.25$
$\sigma^2 = 1$
$\sigma^2 = 4$
n=50
n=100
n=200
n=500
n=1000